

## [I225] Statistical Signal Processing(E) Office Hour 4

1. Consider a continuous-time LTI smoothing filter defined by

$$Y(t) = \frac{1}{2T} \int_{t-T}^{t+T} X(\tau) d\tau.$$

Suppose the input random process  $X(t)$  has power spectral density

$$S_{XX}(\omega) = \frac{1}{1 + \omega^2}.$$

- (a) Find the impulse response  $h(t)$ .
- (b) Find the frequency response  $H(\omega)$ .
- (c) Find the output power spectral density  $S_{YY}(\omega)$ .
- (d) Explain whether the filter behaves as a low-pass or high-pass filter.

Answer:

(a)

The smoothing filter takes the average of the input over the interval  $[t - T, t + T]$ .

Therefore, its impulse response is

$$h(t) = \begin{cases} \frac{1}{2T}, & -T \leq t \leq T, \\ 0, & \text{otherwise.} \end{cases}$$

(b)

The frequency response is obtained by taking the Fourier transform of  $h(t)$ :

$$H(\omega) = \int_{-T}^T \frac{1}{2T} e^{-j\omega t} dt.$$

Thus,

$$H(\omega) = \frac{\sin(T\omega)}{T\omega}.$$

(c)

For an LTI system, the output power spectral density is

$$S_{YY}(\omega) = |H(\omega)|^2 S_{XX}(\omega).$$

Substituting  $H(\omega)$  and  $S_{XX}(\omega)$  gives

$$S_{YY}(\omega) = \frac{\sin^2(T\omega)}{T^2\omega^2} \cdot \frac{1}{1 + \omega^2}.$$

(d)

Since  $|H(\omega)|$  is largest near  $\omega = 0$  and decreases for large  $\omega$ , the filter passes low-frequency components and suppresses high-frequency components. Therefore, it behaves as a low-pass filter.

2. Consider the following first-order differential equation system with white noise input:

$$Y'(t) + 2Y(t) = W(t).$$

The input white noise has autocorrelation

$$R_{WW}(\tau) = 4\delta(\tau).$$

- (a) Find the transfer function  $H(\omega)$ .
- (b) Find the output power spectral density  $S_{YY}(\omega)$ .
- (c) Find the autocorrelation function  $R_{YY}(\tau)$ .
- (d) Determine whether the system is causal or non-causal.

Answer:

(a)

Taking the Fourier transform of the differential equation gives

$$(j\omega + 2)Y(\omega) = W(\omega).$$

Therefore, the transfer function is

$$H(\omega) = \frac{Y(\omega)}{W(\omega)} = \frac{1}{j\omega + 2}.$$

(b)

Because the input is white noise,

$$S_{WW}(\omega) = 4.$$

For an LTI system,

$$S_{YY}(\omega) = |H(\omega)|^2 S_{WW}(\omega).$$

Also,

$$|H(\omega)|^2 = \frac{1}{\omega^2 + 4}.$$

Therefore,

$$S_{YY}(\omega) = \frac{4}{\omega^2 + 4}.$$

(c)

The autocorrelation function is the inverse Fourier transform of the output power spectral density:

$$R_{YY}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{YY}(\omega) e^{j\omega\tau} d\omega.$$

Using the standard Fourier transform pair,

$$\frac{2a}{a^2 + \omega^2} \Leftrightarrow e^{-a|\tau|},$$

with  $a = 2$ , we obtain

$$R_{YY}(\tau) = e^{-2|\tau|}.$$

(d)

The impulse response corresponding to  $H(\omega)$  is

$$h(t) = e^{-2t}u(t).$$

Since  $h(t) = 0$  for  $t < 0$ , the system is causal.

3. You are measuring a constant DC voltage  $\theta$  using a noisy voltmeter. Every time you take a measurement, you get the true value  $\theta$  plus some Gaussian noise  $w$ . You take  $N$  independent measurements:

$$x[n] = \theta + w[n], \quad n = 0, 1, \dots, N-1$$

The noise  $w[n]$  is zero-mean white Gaussian noise where the samples are I.I.D (independent and identically distributed) with a known variance  $\sigma^2$ .

Supposed you decide to estimate the voltage by simply taking the average of all measurement:

$$\hat{\theta} = \frac{1}{N} \sum_{n=0}^{N-1} x[n]$$

(a) Show that this estimator is unbiased.

(b) Compute the variance of  $\hat{\theta}$  and check whether  $\hat{\theta}$  is consistent.

Answer:

(a)

Show that this estimator is unbiased!

Basically, we need to show that  $E\{\hat{\theta}\} = \theta$

$$E\{\hat{\theta}\} = E\left\{\frac{1}{N} \sum_{n=0}^{N-1} x[n]\right\}$$

$$E\{\hat{\theta}\} = \frac{1}{N} \sum_{n=0}^{N-1} E\{\theta + w[n]\}$$

$$E\{\hat{\theta}\} = \frac{1}{N} \sum_{n=0}^{N-1} \theta + 0$$

$$E\{\hat{\theta}\} = \theta$$

(b)

Compute the variance of  $\hat{\theta}$  and check whether  $\hat{\theta}$  is consistent.

$$\text{Var}(\hat{\theta}) = \text{Var}\left(\frac{1}{N} \sum_{n=0}^{N-1} x[n]\right) \boxed{\text{Apply variance operation}}$$

$$\text{Var}(\hat{\theta}) = \frac{1}{N^2} \text{Var} \left( \sum_{n=0}^{N-1} x[n] \right) \quad \boxed{\text{Square Rule: taking constant outside of Var}(\cdot) \text{ will square it.}}$$

$$\text{Var}(\hat{\theta}) = \frac{1}{N^2} \sum_{n=0}^{N-1} \text{Var}(x[n]) \quad \boxed{\text{Independence rule}}$$

$$\text{Var}(\hat{\theta}) = \frac{1}{N^2} \sum_{n=0}^{N-1} \sigma^2 \quad \boxed{\text{Var}(x[n]) = \text{Var}(\theta + w[n]) = \text{Var}(w[n]) = \sigma^2}$$

$$\text{Var}(\hat{\theta}) = \frac{\sigma^2}{N}$$

An estimator is consistent if it is unbiased and its variance goes to zero as  $N \rightarrow \infty$ .

$$\lim_{N \rightarrow \infty} \text{Var}(\hat{\theta}) = \lim_{N \rightarrow \infty} \frac{\sigma^2}{N} = 0$$

Conclusion:  $\hat{\theta}$  is consistent.

4. A sensor measures the distance to a target  $x[n]$ . Due to a specific type of quantization error, the measurements are uniformly distributed between 0 and an unknown maximum distance  $\theta$ :

$$p(x; \theta) = \begin{cases} \frac{1}{\theta}, & 0 \leq x \leq \theta \\ 0, & \text{otherwise} \end{cases}$$

Find the Cramér-Rao Lower Bound (CRLB) of  $\theta$ , if it is a valid boundary.

Answer:

Check the regularity conditions.

Before finding the CRLB, we first need to show that regularity conditions have been fulfilled:

$$E \left\{ \frac{\partial \ln p(x; \lambda)}{\partial \lambda} \right\} = 0$$

Below is the step-by-step:

$$p(x; \theta) = \frac{1}{\theta}$$

$$\ln p(x; \theta) = -\ln \theta$$

$$\frac{\partial \ln p(x; \theta)}{\partial \theta} = -\frac{1}{\theta}$$

$$E \left\{ \frac{\partial \ln p(x; \theta)}{\partial \theta} \right\} = E \left\{ -\frac{1}{\theta} \right\} \neq 0$$

Since regularity conditions are not fulfilled, using CRLB is not valid for this.

5. A sensor counts the number of packets  $x[n]$  arriving at a network switch every second. The counts are independent and follow a Poisson distribution with an unknown arrival rate  $\lambda$ :

$$p(x[n]; \lambda) = \frac{e^{-\lambda} \lambda^{x[n]}}{x[n]!}$$

After some time,  $N$  independent counts have been observed.

Find the Cramér-Rao Lower Bound (CRLB) of  $\lambda$ , if it is a valid boundary.

Answer:

Note: Since there are  $N$  observations that have been done, we are dealing with the joint PDF, not the individual PDF:

$$p(\mathbf{x}; \lambda) = \prod_{n=0}^{N-1} \frac{e^{-\lambda} \lambda^{x[n]}}{x[n]!}, \quad \mathbf{x} = [x[0], x[1], \dots, x[N-1]]^T$$

(a)

Check the regularity condition.

Before finding the CRLB, we first need to show that regularity conditions have been fulfilled:

$$E \left\{ \frac{\partial \ln p(\mathbf{x}; \lambda)}{\partial \lambda} \right\} = 0$$

Below is the step-by-step:

$$p(\mathbf{x}; \lambda) = \prod_{n=0}^{N-1} \frac{e^{-\lambda} \lambda^{x[n]}}{x[n]!}$$

$$\ln p(\mathbf{x}; \lambda) = \sum_{n=0}^{N-1} -\lambda + x[n] \ln \lambda - \ln (x[n]!)$$

$$\frac{\partial}{\partial \lambda} \ln p(\mathbf{x}; \lambda) = \sum_{n=0}^{N-1} -1 + \frac{x[n]}{\lambda} - 0$$

$$E \left\{ \frac{\partial}{\partial \lambda} \ln p(\mathbf{x}; \lambda) \right\} = E \left\{ \sum_{n=0}^{N-1} -1 + \frac{x[n]}{\lambda} \right\}$$

$$E \left\{ \frac{\partial}{\partial \lambda} \ln p(\mathbf{x}; \lambda) \right\} = \sum_{n=0}^{N-1} -1 + E \left\{ \frac{x[n]}{\lambda} \right\}$$

$$E \left\{ \frac{\partial}{\partial \lambda} \ln p(x; \lambda) \right\} = \sum_{n=0}^{N-1} -1 + \frac{1}{\lambda}(\lambda) = 0$$

From equations above, it is proven that regularity condition have been fulfilled.

(b)

Calculate Fisher's Information  $I(\lambda) = -E \left\{ \frac{\partial^2 \ln p(x; \lambda)}{\partial \lambda^2} \right\}$

$$\frac{\partial^2 \ln p(x; \lambda)}{\partial \lambda^2} = \frac{\partial}{\partial \lambda} \left( \frac{\partial}{\partial \lambda} \ln p(x; \lambda) \right) = \frac{\partial}{\partial \lambda} \left( \sum_{n=0}^{N-1} -1 + \frac{x[n]}{\lambda} \right) = \sum_{n=0}^{N-1} 0 + \frac{-x[n]}{\lambda^2}$$

$$I(\lambda) = -E \left\{ \frac{\partial^2 \ln p(x; \lambda)}{\partial \lambda^2} \right\} = -E \left\{ \sum_{n=0}^{N-1} \frac{-x[n]}{\lambda^2} \right\} = -\sum_{n=0}^{N-1} \frac{-E\{x[n]\}}{\lambda^2} = \frac{N}{\lambda}$$

(c)

Calculate CRLB.

From Fisher's Information, we can conclude that any unbiased estimator  $\hat{\lambda}$  satisfies

$$\text{Var}(\hat{\lambda}) \geq \frac{\lambda}{N}$$